

3D Coordinates Extension — Build the Spider (Beginner)

Topic: 3D Spider (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner (Extension, after Lesson 6) · **Time:** about 30 minutes

Build a red spider in 3D — **red body, eight legs, two eyes**. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

- Stand on your **home spot** (red block) every run. Move, and the spider moves too.
- 👁 Red parts = **red concrete**. Eyes = **black wool**.

1 · Predict 🧠

Read the code. Circle one word.

```
place red block at (1, 1, 2)
place red block at (1, 1, 3)
place red block at (1, 1, 4)
```

Only z changes. This leg steps... Circle one: across · up · deeper

```
place red block at (1, 1, 2)
place red block at (5, 1, 2)
```

Only x changes. This jumps from the left side to the right side. Circle one: across · up · deeper

2 · Spot the Bug 🐛

An eye sits **up high on the front** of the body, at (2, 3, 3). The code below has only **two** numbers, so it drops flat on the floor.

```
# broken  
place black block at (2, 3)
```

Fill the blank to fix it:

```
place black block at (2, 3, __)
```

Which number fills the blank? Circle one: 3 · 9

What was missing? Circle one: the z number · the color

3 · Show Your Work 📷 🎥

Look at the 🕷️ spider in the world. Walk around it. See the fat body, the eight legs spread out, and the two eyes on the front.

Build a small spider on your home spot. Keep it easy:

Red body: fill red blocks from (2, 2, 3) to (4, 3, 4) — a small block.

Four legs on the left: red block at (1, 1, 2), (1, 1, 3), (1, 1, 4), (1, 1, 5).

Four legs on the right: red block at (5, 1, 2), (5, 1, 3), (5, 1, 4), (5, 1, 5).

Two black eyes on the front: black block at (2, 3, 3) and (4, 3, 3).

💡 Tip: legs stay **low** (small y). Left legs all share $x = 1$, right legs all share $x = 5$. Each leg steps **deeper** (bigger z).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the spider. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.